[Total No. of Question -04] [Total No. of Pages - 01]

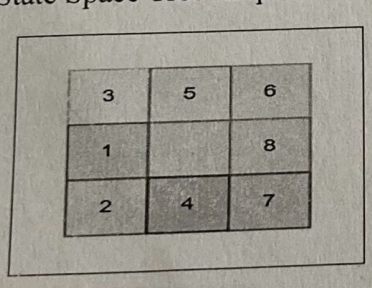
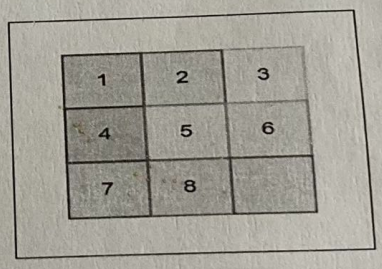
G H Raisoni International Skill Tech University, Pune

S.Y. B.Tech (CE/IT/CS/AIML/AIDS/AI)
(TERM-III)
CAE II- WINTER 2025 (2024 Pattern)
Introduction to Artificial Intelligence (25BTPCL309)

(Time: - 01 Hour) (Max. Marks: 20)

Instructions to the students:

i) All questions are compulsory. ii) Neat diagrams must be drawn wherever necessary.
iii) Figures to the right indicate full marks. iv) Use of scientific calculator is allowed
v) Assume suitable data, if necessary.

Q.No	Question	Marks	CO	BL
Q1.(a)	Explain how Constraint Satisfaction technique is applied to Solve Map Coloring Problem with an example.	[5]	3	2
b)	Describe the Concept Describe and Match Technique with Medical Diagnosis case study.	[5]	3	3
	OR			
Q 2.(a)	Illustrate how Goal Reduction Technique can be used to solve real time Traffic problem of Pune. Give AI Solution and Explain with diagram.	[5]	3	2
(b)	Describe importance of Neural Network with its Architecture and real time applications in detail.	[5]	4	3
Q 3.(a)	Apply Heuristic Method (A* Algorithm) to solve 8 Puzzle problem. Draw State Space Tree .Explain in detail. Consider following Initial and Goal State.  Start state  Goal state	[5]	3	3
(b)	Explain DFS and BFS search techniques with example.	[5]	3	
	OR			
Q. 4 (a)	Illustrate Rule based Concept in AI with real time case study.	[5]	3	
(b)	Explain Ontology with real time case study from Automobile Industry.	[5]	3	